

Erratum

Egfr signaling regulates ommatidial rotation and cell motility in the *Drosophila* eye via MAPK/Pnt signaling and the Ras effector Canoe/AF6

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An error in this article was not corrected before going to press.

In Fig. 5, the arrows between Ras^{V12C40} and PI3K, and Ras^{V12C40} and Rgl were missing in panels D and F.

We apologise to readers and the authors for this mistake.

Egfr signaling regulates ommatidial rotation and cell motility in the *Drosophila* eye via MAPK/Pnt signaling and the Ras effector Canoe/AF6

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Summary

Epidermal Growth Factor-receptor (Egfr) signaling is evolutionarily conserved and controls a variety of different cellular processes. In *Drosophila* these include proliferation, patterning, cell-fate determination, migration and survival. Here we provide evidence for a new role of Egfr signaling in controlling ommatidial rotation during planar cell polarity (PCP) establishment in the *Drosophila* eye. Although the signaling pathways involved in PCP establishment and photoreceptor cell-type specification are beginning to be unraveled, very little is known about the associated 90° rotation process. One of the few rotation-specific mutations known is *roulette* (*rlt*) in which ommatidia rotate to a random degree, often more than 90°. Here we show that *rlt* is a rotation-specific allele of the inhibitory Egfr ligand Argos and that modulation of

Egfr activity shows defects in ommatidial rotation. Our data indicate that, beside the Raf/MAPK cascade, the Ras effector Canoe/AF6 acts downstream of Egfr/Ras and provides a link from Egfr to cytoskeletal elements in this developmentally regulated cell motility process. We provide further evidence for an involvement of cadherins and non-muscle myosin II as downstream components controlling rotation. In particular, the involvement of the cadherin Flamingo, a PCP gene, downstream of Egfr signaling provides the first link between PCP establishment and the Egfr pathway.

Key words: Canoe/AF6, Egfr, Ommatidial rotation, Planar cell polarity (PCP), *roulette* (*rlt*), Argos, Flamingo

Introduction

Epithelial planar cell polarity (PCP) in the *Drosophila* eye is generated when immature ommatidial preclusters acquire opposite chirality in the dorsal and ventral halves of the eye imaginal disc and subsequently rotate 90° towards the dorsoventral midline, the equator (Adler, 2002; Mlodzik, 1999; Mlodzik, 2002; Reifegerste and Moses, 1999). The signaling pathways that regulate the establishment of ommatidial chirality are beginning to be understood and require the specification of distinct cell fates within the R3/R4 photoreceptor pair. This is generated through an interplay of Frizzled (Fz)/PCP and Notch signaling. Fz/PCP signaling induces R3 and leads to the expression of the Notch ligand Delta in R3. Delta subsequently activates Notch in the neighboring cell, specifying it as R4 (Cooper and Bray, 1999; Fanto and Mlodzik, 1999; Tomlinson and Struhl, 1999). The direction of the subsequent rotation follows the cell-type specification of the R3/R4 photoreceptors (Adler, 2002; Mlodzik, 2002). However, in *fz* and other PCP mutants as well as in *Notch* the mechanistic aspects of rotation are not abolished (Adler, 2002; Mlodzik, 1999; Mlodzik, 2002; Reifegerste and Moses, 1999), suggesting that other signaling pathways and mechanisms regulate the cell motility aspects of rotation.

During this process each ommatidium rotates 90° towards

the equator and thus in opposite directions in the dorsal and ventral halves. This morphogenetic event is remarkable, as it requires groups of cells to undergo a coordinated precise movement: cells of each ommatidial precluster stick together and move as a unit with respect to the surrounding epithelial cells (Mlodzik, 1999; Reifegerste and Moses, 1999). The rotation of ommatidial preclusters is initiated in the third instar eye imaginal disc, shortly after the Fz/PCP-Notch signaling interplay specifies cell fate, and is visibly appreciated around column 6 posterior to the morphogenetic furrow (see Fig. 1A-C). It is completed by column 15-18 approximately 20-24 hours later. During this process, the rotation angles of individual ommatidia can be visualized with markers that highlight either the R3/R4 precursor pair or the R1/R6 pair. Interestingly, ommatidial rotation appears to be a two-step process as clusters stop for 3-4 columns after undergoing the first 45° rotation and before initiating the second 45° to complete the 90° rotation. In the adult eye the rotation is represented in the precise mirror image arrangement of the two chiral ommatidial forms that face each other on opposite sites of the equator (see Fig. 1D,E).

Very little is known about the regulation and mechanistic aspects of ommatidial rotation as only a few mutants that specifically affect this process have been isolated. In such 'rotation specific' mutations both the direction of rotation and

associated chirality-R3/R4 specification are unaffected, whereas the degree of rotation of individual ommatidial clusters is abnormal. Mutations falling into this category include *nemo* (*nmo*) (Choi and Benzer, 1994), *roulette* (*rlt*) (Choi and Benzer, 1994), *rok* (the *Drosophila* Rho-kinase) (Winter et al., 2001) and *scabrous* (*sca*) (Chou and Chien, 2002).

The requirement of *rok* in ommatidial rotation provides the only link to the Fz/PCP pathway as Rho-kinase acts downstream of RhoA in several cellular contexts. Particularly in the *Drosophila* wing, *rok* has been shown to act downstream of Fz/Dishevelled and RhoA in PCP establishment, affecting wing hair number (2-3 cell) but not their orientation (Winter et al., 2001). *rok* has been reported to affect rotation (and photoreceptor number), but not the R3/R4 cell fate decision (which is equivalent to wing hair orientation). This might suggest that some aspects of wing hair growth and rotation are shared through similar regulatory input into cytoskeletal regulation.

The *nmo* gene encodes a distant member of the Map kinase family, and in *nmo* mutants most ommatidia arrest at a rotation angle of approximately 45°, failing to execute the second 45° turn (Choi and Benzer, 1994). In *sca*, mutant clusters generally rotate more than 90° (Chou and Chien, 2002). In contrast, *rlt* mutant ommatidia rotate both less and more than the normal 90° (approximately 50% each) (see Fig. 1F), suggesting a general regulatory role in this process. The observation that *nmo* is epistatic to *rlt* and that *nmo*, *rlt* double mutants show the *nmo* phenotype (with ommatidia arresting at around 45°), lead to the hypothesis that *rlt* is required to stop rotation (Choi and Benzer, 1994).

Here we show that *rlt* is a rotation-specific allele of the Egfr-inhibitory ligand Argos. Both reduction and increase in Egfr signaling lead to defects in ommatidial rotation, suggesting that Egfr signaling is generally required to regulate this process. Using a combination of mutant analysis, genetic interactions and specific Ras-effector loop mutations we have identified both the Raf/MAPK cascade and the novel Ras effector Canoe/AF6 as a mediator of Egfr/Ras signaling in this process (Kuriyama et al., 1996). Our data indicate that Canoe/AF6 provides a link from Egfr to cytoskeletal elements in this developmentally regulated cell motility process. Furthermore, we provide evidence that Egfr signaling acts on cell adhesion via effects through the cadherin Flamingo, thus providing a link between Egfr and the PCP genes.

Materials and methods

Fly stocks

If not otherwise mentioned, fly stocks are as described in Flybase (<http://flybase.bio.indiana.edu/>), provided by The Bloomington Stock Center (<http://flystocks.bio.indiana.edu/>). The wild-type (WT) stock used was *OregonR*. *rlt¹* and *nemo^{P1}* mutants were kindly provided by K.W. Choi (Choi and Benzer, 1994). The *argos^{W11}*, *argos⁴⁷* flies (Freeman et al., 1992) were kindly provided by C. Klambt and *sev-argos* by M. Freeman (Freeman, 1994). Deficiencies used that failed to complement the *rlt* rough eye phenotype were: *Df(3L)st7P*, *Df(3L)st-j7*, *Df(3L)st-f13*, *Df(3L)81k19*, *Df(3L)st7*, whereas the adjacent deficiencies *Df(3L)D-5rv12*, *Df(3L)th102*, *Df(3L)std11*, *Df(3L)w10*, *Df(3L)cat* and deficiencies more proximal or distal to the 70C-75F interval did complement *rlt*.

The Egfr alleles *top¹* and *Egfr^{EC20}* (Clifford and Schupbach, 1989),

as well as the *S⁴⁸⁻⁵* allele and *UAS-λ-top* (Queenan et al., 1997) were kindly provided by T. Schupbach. *UAS-PI3K* and *Dp110A* flies were from Sally Leever (Leever et al., 1996). The *GMR-Ral^{G23V}* flies are as published (Sawamoto et al., 1999). The *cno^{Mis1}* allele (Miyamoto et al., 1995) was provided by D. Yamamoto. *UAS-cno*, *cno²*, *cno³* (Matsuo et al., 1997) and the *UAS-RasV12* isoforms (Karim and Rubin, 1998) were kindly provided by U. Gaul. The signaling specificity of the *RasV12* effector loop isoforms was confirmed in *Drosophila* (Karim and Rubin, 1998; Prober and Edgar, 2002).

Embedding and sectioning of eyes was performed as described (Tomlinson, 1987). Crosses were performed at 25°C (except for *aos^{w11}* at 18°C). Multiple eyes were sectioned and analyzed for each genotype.

Molecular characterization of the *rlt* mutation

The hypomorphic behavior of the *rlt* allele and the absence of any other obvious phenotype led us to the conclusion that *rlt* is probably a regulatory mutant of *aos*. As regulatory mutations often affect untranslated regions of a gene, whereas the protein is not altered, we analyzed the genomic *aos* region in homozygous *rlt* flies using overlapping PCR primers. Two of the overlapping primer sets gave PCR products that were approximately 1 kb larger than the WT control. Sequencing of these PCR products revealed a truncated P-element inserted in the 5' UTR of *aos* within the proposed transcriptional start site at bp 55 according to the published *aos* sequence M91381. Primers used to amplify the truncated P-element from genomic DNA were: 5' CACAGACACGCACATACCG 3' and 5' CCCTCGCTCTATCGTTGTTC 3'.

Generation of the *mδ0.5-Gal4* construct

The *mδ0.5-Gal4* construct was generated by cloning a *XhoI-EcoRI* *mδ0.5*-fragment (PCR amplified from a plasmid, kindly provided by S. Bray using primers: 5'-CCGCTCGAGTGCCATCAGATGTC-AGCAAATG-3' and 5'-CGGAATTCCTTTTGGCGCACAGTCA-CAC-3') upstream of an *EcoRI-BamHI* fragment containing the P-element minimal-promoter (PCR amplified from pCasper 4 using the primers: 5'-CGGAATTCAAAGCCGAAGCTTACCGAAGT-3' and 5'-CGGGATCCTTTTTTTTATTCCACGTAAGG-3').

The *Gal4* gene was added as a *BamHI-NotI* fragment (from pGaTB). The final construct was cloned as an *Asp718-NotI* fragment from pBSSK⁺ into pCasper4. Several fly lines were established according to standard procedures. *mδ0.5-Gal4* drives expression mainly in R4 (and also weaker early in R3 and later in R7), as confirmed by ablation experiments and expression of an *UAS-GFP* transgene.

Immunofluorescence

Imaginal discs were dissected, fixed and stained as described (Freeman et al., 1992). Antibodies used were: rat anti-Elav (1:50), mouse anti-Boss (1:1000) (from Developmental Studies Hybridoma Bank), rabbit anti-Bar (1:100) (a generous gift from K. Saigo) (Higashijima et al., 1992), mouse anti-Fmi (1:10) (gift from Tadashi Uemura), rat anti-DE-cadherin (DCAD2, 1:20) (gift from H. Oda) (Oda et al., 1994) and mouse (Promega) or rabbit anti-βGal (Molecular Probes) (1:2000). Images were acquired on a Leica TCS-SP (UV) confocal microscope and processed in Adobe Photoshop.

Results

rlt is a rotation-specific allele of *argos*

We mapped *rlt*, originally described as a spontaneous mutation on chromosome III (Choi and Benzer, 1994), by complementation analysis with overlapping deficiencies to the 72D10-73B1 interval on chromosome 3L (not shown). Complementation analysis with candidate genes within this

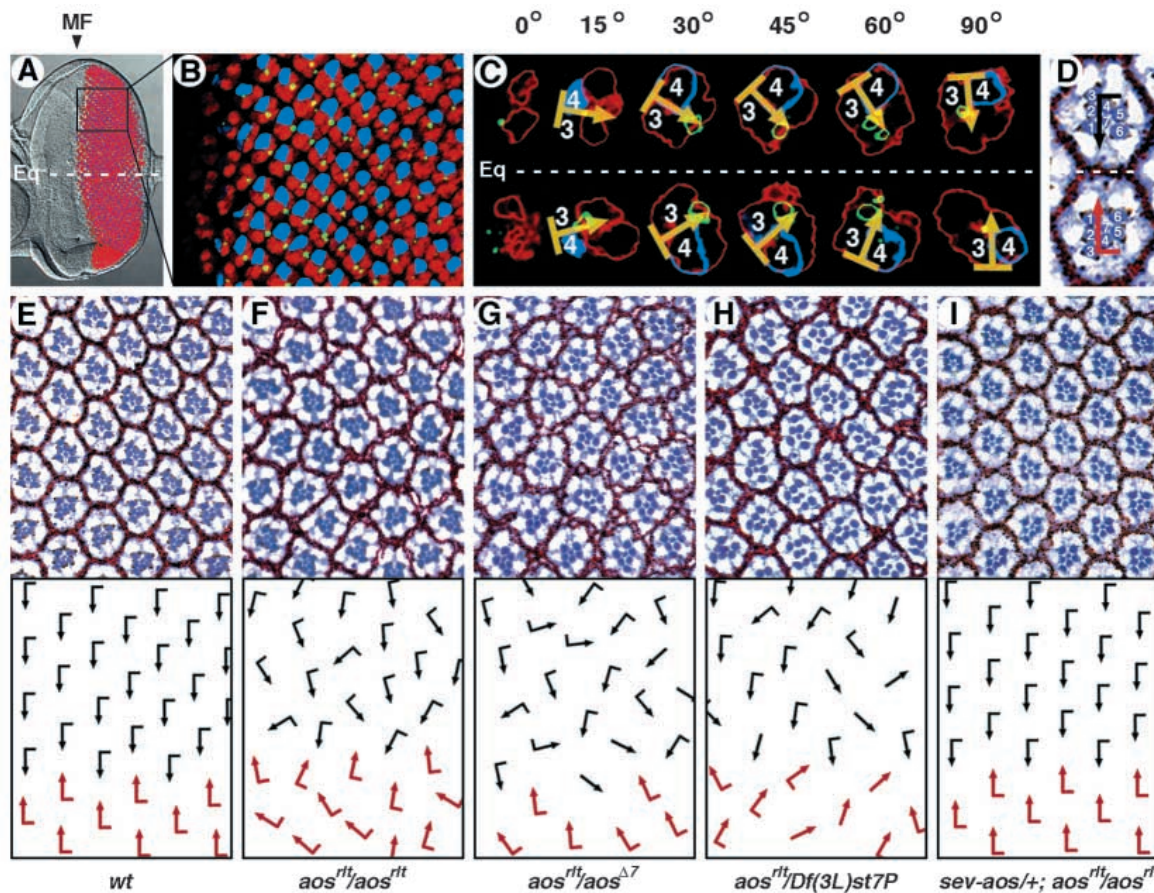


Fig. 1. The *roulette* mutant is an allele of the Egfr-ligand *argos*. Panels show eye imaginal discs (A-C) or adult eye sections (D-I). Anterior is left and dorsal up. (A) Eye-imaginal disc stained with anti-Elav (red, all photoreceptor precursors) and *m80.5-lacZ* (blue, R4 precursors) shown as an overlay with a Nomarski optics bright-field view of same disc. The dorsoventral midline, equator (Eq), is marked by white broken line, the position of the morphogenetic furrow (MF) is indicated. (B) Higher magnification of boxed area from A also showing anti-Boss (green, marking the central R8 cell in each cluster). Note regular arrangement and orientation of ommatidial clusters. Overlay of the red and blue channels is for clarity shown in blue. (C) Semi-schematic presentation of ommatidial rotation (generated from real clusters shown in A,B with the 'solarize' filter of Adobe Photoshop). Representative clusters for the dorsal and ventral halves are shown. Colors are as in B. R3/R4 photoreceptor precursors are indicated by white numbers. Ommatidial preclusters represent examples from columns 4, 5, 7, 9, 12 and 15, respectively. Yellow arrows indicate rotation of clusters. The rotation angle of individual clusters is indicated above the panel. The 'base' of arrows indicates the R3/R4 axis and the arrowheads point towards the middle of cluster. Note successive changes in the rotation angles to the complete 90°. (D) Examples of mature adult ommatidia with numbered photoreceptor rhabdomeres (R1-R7; R8 is below the plane of section) and schematic arrows as shown in E-I. Black arrow: dorsal ommatidium; red arrow: ventral ommatidium. (E-I) Tangential sections of equatorial region of adult eyes with schematic presentations below sections. (E) *wild-type* (*wt*), (F) *argos-roulette* (*aos^{rlt}/aos^{rlt}*), (G) *aos^{rlt}/aos^{Δ7}*, (H) *aos^{rlt}/Df(3L)st7P*, (I) *sev-aos/+; aos^{rlt}/aos^{rlt}*. Note that *aos^{rlt}* fails to complement *aos* alleles and that *sev-aos* fully rescues the *aos^{rlt}* phenotype. Ommatidia with a *wt* photoreceptor number are represented by arrows with flags; clusters with extra photoreceptors are indicated by simple arrows.

region revealed that alleles of the secreted Egfr inhibitory ligand *argos* (Freeman et al., 1992) failed to complement *rlt* (Fig. 1G). Transheterozygous *rlt/aos* and *rlt/Df(3L)st7P(aos⁻)* flies displayed a very similar phenotype with rotation defects resembling homozygous *rlt* flies (Fig. 1G,H). In addition, the extra photoreceptor cell phenotype reported in a small number of *rlt* ommatidia (Choi and Benzer, 1994) is enhanced in *rlt/aos* and *rlt/Df(3L)st7P(aos⁻)* combinations (Fig. 1G,H). This is intriguing, as the most prominent phenotype of *aos* null clones and strong hypomorphic alleles are 1-2 extra photoreceptors per ommatidium caused by a lack of Egfr inhibition (Freeman et al., 1992). Thus, in complementation analysis *rlt* behaves as a hypomorphic allele of *aos*.

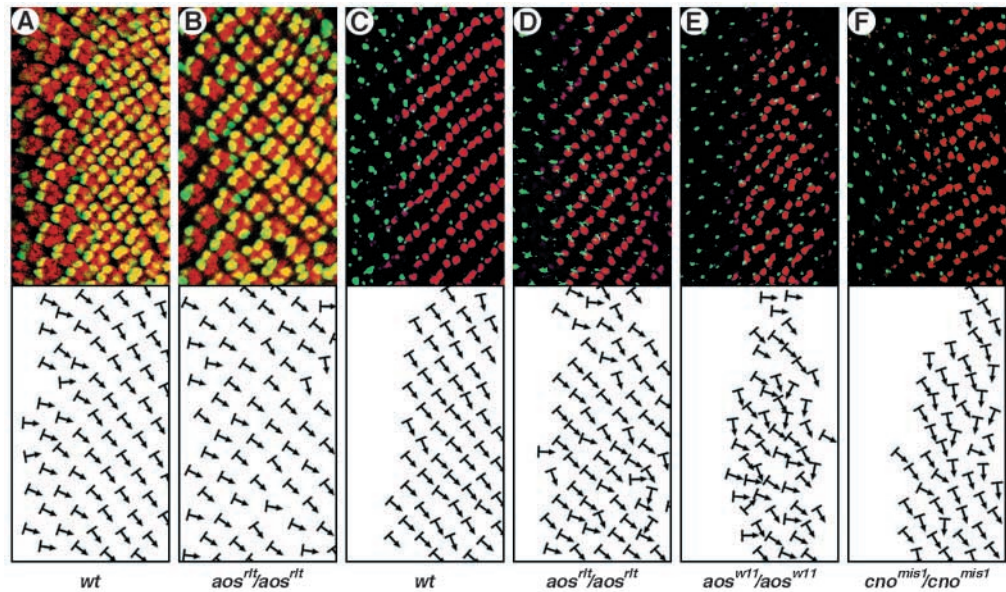
To confirm these results we attempted to rescue *rlt* with an *argos* transgene. The *sev-argos* transgene rescues the *argos*

loss-of-function (LOF) eye phenotype and has no dominant effect on eye development (Freeman, 1994). Strikingly, *sev-argos* completely rescued the *rlt* eye phenotype to WT (Fig. 1I). These data indicated that *rlt* is a hypomorphic, rotation-specific allele of *aos*. In addition, molecular analysis of *aos^{rlt}* revealed an insertion of a truncated P-element in the 5' untranslated region within the proposed transcription start site of *aos* (see Materials and methods for details), indicating that *rlt* is most probably a regulatory mutation of *aos*. We will subsequently refer to the *rlt* allele as *aos^{rlt}*.

rlt and *argos* discs exhibit rotation defects

To refine the phenotypic features of *aos^{rlt}* we have analyzed *aos^{rlt}* eye imaginal discs using markers that highlight cell-type identity and orientation of each cluster at the time when

Fig. 2. Eye imaginal discs mutant for *argos* or *canoe* exhibit defects in ommatidial rotation. Panels show a dorsal field of third instar eye imaginal discs, anterior is left and dorsal up. Discs in A,B are double-stained with anti-Elav (red; all photoreceptors) and *svp-lacZ* (green; marking R3/R4 and R1/R6). (C-F) Double-labeled with anti-Bar (red; labeling R1/R6) and anti-Boss (green; marking R8). With these markers rotation angles can be unambiguously determined and are schematically shown with arrows below each panel (arrows are drawn according to Fig. 1C). (A,C) *wt*-type, (B,D) *aos^{rlt}/aos^{rlt}*, (E) *aos^{w11}/aos^{w11}*. Note misrotations in both *aos* alleles. (F) *cno^{mis1}/cno^{mis1}*. Discs were co-stained with anti-Elav to ensure that rotation abnormalities are not because of photoreceptor loss (blue channel, not shown).



chirality is being established and ommatidial rotation takes place (Fig. 2 and not shown). Whereas in WT discs a fairly regular pattern is observed with all markers (Fig. 2A,C and not shown; see Figure legend for markers), in *aos^{rlt}* discs many clusters reveal rotation abnormalities (Fig. 2B,D). We observe clusters that are both overrotated and underrotated with respect to their neighbors and developmental stage (Fig. 2B,D). Discs of the strong *aos^{w11}* allele showed, in addition to the extra R-cell phenotype, even more severe rotation defects (Fig. 2E), again indicating that *aos* is generally required during ommatidial rotation. Based on the relatively regular appearance of the first *svp-lacZ* and Bar-positive rows (Fig. 2B,D), it appears that the initial 45° rotation step is less affected.

We found no evidence for an *aos* effect on the R3/R4 cell-fate decision and ommatidial chirality, as the expression of the *m80.5lacZ* marker, highlighting the R4 photoreceptor, appears normal within the R3/R4 pair (not shown; see below). Thus we conclude that *argos* is generally required for the process of ommatidial rotation, but has no effect on chirality and associated R3/R4 cell-fate decisions.

Ommatidial rotation is controlled by Egfr signaling

The requirement of Argos as an inhibitory ligand of Egfr (Jin et al., 2000) and the identification of *argos^{rlt}* suggested that Egfr signaling plays a critical role in ommatidial rotation. As *Egfr* is very pleiotropic with multiple requirements throughout development and eye patterning in particular (Bogdan and Klambt, 2001; Casci and Freeman, 1999; Freeman, 1997; Schweitzer and Shilo, 1997; Van Buskirk and Schupbach, 1999), it is not possible to analyze the role of *Egfr* in ommatidial rotation using mitotic clones of null alleles. However, hypomorphic alleles of *Egfr* allow the analysis of rotation defects. Eyes of the *Egfr^{top1}* allele (Clifford and Schupbach, 1989), for example, are mildly rough and reveal, beside the expected photoreceptor loss (Freeman, 1997), many ommatidia with rotation defects (Fig. 3A). Similarly, rotation is affected in hetero-allelic *Egfr^{top1}/Egfr^{E20}* combinations (Fig. 3B).

Further evidence for an involvement of Egfr signaling in

ommatidial rotation came from the analysis of *Star* (*S*) mutants. *Star* is a chaperone required for processing of the secreted Egfr ligand Spitz (Lee et al., 2001), and Egfr signaling is thus reduced in *S* mutants. *S* is haplo-insufficient for eye development and loss of one gene copy leads to a rough eye phenotype because of the reduction in Egfr activation (Kolodkin et al., 1994). *S^{-/+}* eyes show a very mild loss of photoreceptors and, in addition, rotation abnormalities (Fig. 3C). Interestingly, the *S^{-/+}* rotation phenotype is enhanced when one copy of *Egfr* or *Ras* is removed (Fig. 3D,E, Table 1), and is suppressed by *aos^{rlt}* heterozygosity (not shown). In addition, we see a specific dominant enhancement of the *S^{-/+}* rotation phenotype by the removal of one copy of *spitz*, whereas the other known Egfr ligands *vein* and *gurken* had no effect (Table 1), suggesting that Spitz is the activating Egfr ligand in this context. The recently described Egfr ligand Keren could not be tested, because no mutants are available (Reich and Shilo, 2002). Taken together, these observations indicate that Egfr/Ras signaling is directly involved in regulating ommatidial rotation, a requirement that is masked by other effects seen in null clones or severe hypomorphic mutations.

Egfr affects rotation through the Ras/MAPK/Pointed cascade

Because loss of the inhibitory ligand Aos results in gain of Egfr signaling, we asked if reduction of Egfr signaling components could suppress the rotation defects in *aos^{rlt}*. Given that the *argos* rough eye phenotype is dosage-sensitive to Egfr-signaling components (Sawamoto et al., 1996), we analyzed *aos^{rlt}* flies in a heterozygous background of *Egfr* or its downstream effectors to obtain an insight into the effector pathways employed in ommatidial rotation. We found that the rotation defects in *aos^{rlt}* eyes were suppressed in an *Egfr^{top1/+}* or *Ras^{e2F/+}* and *pnt^{Δ88/+}* background (Fig. 4), arguing for a requirement of the conserved Ras/MAPK/Pnt cascade in rotation downstream of Egfr. This is further supported by the fact that *aos* is a direct transcriptional target of the ETS

Table 1. Genetic interactions with the *Star* haploinsufficient rough eye phenotype

Gene (<i>S-48.5/+</i> for all genotypes below)	% misrotated ommatidia $\pm$ s.d.	Molecular features of gene
Control (<i>Ore-R</i>)	9.55 $\pm$ 1.43	
Egfr ligands		
<i>spi-259</i>	37.83$\pm$13.35	Activating Egfr ligand
<i>spi-1</i>	40.04$\pm$3.70	Activating Egfr ligand
<i>grk-HK36</i>	3.14 $\pm$ 2.58	Activating Egfr ligand
<i>vein-addC6</i>	10.02 $\pm$ 3.27	Activating Egfr ligand
<i>aos-rlt</i>	3.74 $\pm$ 1.99	Inhibitory Egfr ligand
'Signaling factors'		
<i>Egfr-top1</i>	41.52$\pm$7.24	Egfr
<i>Ras-e2F</i>	58.05$\pm$4.57	Ras-GTPase
<i>cno-2</i>	40.56$\pm$7.99	Ras-effector, junction-binding
<i>cno-3</i>	43.24$\pm$8.67	Ras-effector, junction-binding
<i>nmo-P1</i>	17.38 $\pm$ 5.39	Mapk family member
Cell adhesion factors		
<i>shg-2</i>	40.56$\pm$5.21	DE-cadherin
<i>arm-XM19</i>	12.43 $\pm$ 5.66	β -Catenin
<i>fmi-E59</i>	45.59$\pm$7.32	Atypical Cadherin
<i>wb-09437</i>	13.20 $\pm$ 3.05	Laminin
<i>wb-4418</i>	7.17 $\pm$ 0.60	Laminin
<i>mys-null</i>	4.44 $\pm$ 2.59	Integrin β -subunit
<i>mys-1</i>	7.64 $\pm$ 3.82	Integrin β -subunit
Other		
<i>enabled-02029</i>	7.77 $\pm$ 2.95	VASP homolog, adaptor protein
<i>peanut-XP</i>	24.58$\pm$5.90	Septin
<i>peanut-02502</i>	20.07$\pm$11.25	Septin
<i>zipper-02957</i>	34.60$\pm$4.71	Non-muscle Myosin II
<i>zipper-1</i>	19.34$\pm$7.06	Non-muscle Myosin II
<i>khc-8</i>	10.38 $\pm$ 5.72	Kinesin heavy chain
<i>khc-k13314</i>	2.87 $\pm$ 2.04	Kinesin heavy chain
<i>ck-13</i>	5.41 $\pm$ 2.22	Myosin VIIa
<i>capt-06955</i>	34.19$\pm$8.94	Adenylyl cyclase-associated protein
<i>karst-01318</i>	5.62 $\pm$ 1.13	β -Heavy spectrin
<i>sca-BP1</i>	9.75 $\pm$ 3.06	Secreted factor
<i>chic-01320</i>	21.90$\pm$5.98	Profilin

All interactions are in heterozygous genotypes of the indicated gene. For all genotypes, *n* is at least 300 ommatidia from three to six independent eyes, the standard deviation (s.d.) is between eyes. Genotypes and numbers shown in bold represent genes that show a significant enhancement of the *Star*^{+/+} rotation phenotype. Only normally constructed ommatidia were considered (ommatidia with an abnormal photoreceptor complement do not contribute to the total% of ommatidia analyzed). For detailed information of the alleles used see also <http://flybase.bio.indiana.edu/>

domain transcription factor Pointed (*pnt*), which is itself activated by the MAPK Rolled (Golembo et al., 1996). Surprisingly, removing one copy of *raf* and *rl/Mapk* did not suppress the rotation defects of *aos^{rlt}* (not shown), suggesting that these kinases are not rate limiting in this process. Nevertheless, the facts that *pnt* acts as the strongest suppressor of *aos^{rlt}* and that Pnt is activated by *RI/Mapk* argue for a requirement of the conserved Ras/MAPK/Pnt pathway in rotation. In addition, removal of one copy of the *nmo* gene also suppresses the *aos^{rlt}* phenotype (Fig. 4E), suggesting a link between Egfr signaling and *nmo*. This is in line with previous observations proposing that *nmo* is epistatic to *rlt*: the *nmo, rlt* double mutant exhibits a *nmo* phenotype (Choi and Benzer, 1994).

Ras-effector loop mutants identify additional Egfr effectors in rotation

As ommatidial rotation is a cell motility process requiring cytoskeletal rearrangements, we wondered if other effectors of Egfr beside the Raf/MAPK cascade play a role in this process. The Ras GTPase, the main transducer of Egfr signaling, can utilize distinct effectors in different contexts. In addition to nuclear signaling, mediated by the Raf/MAPK/Pnt cascade, Ras can affect cell growth and cytoskeletal rearrangements via its effectors Rgl/Ral, Phospho-inositol-3-Kinase (PI3K) and Canoe, whose human homologue (AF6) is known as the critical partner of ALL1 in a chimeric protein associated with myeloid leukemia (Joneson et al., 1996; Karim and Rubin, 1998; Kuriyama et al., 1996; Prasad et al., 1993; Prober and Edgar, 2002; Rodriguez-Viciano et al., 1997).

Several point mutations have been identified within the Ras-effector loop (amino acids 34–41) that abrogate the binding to and activation by Ras of specific effectors (Joneson et al., 1996; Karim and Rubin, 1998; Rodriguez-Viciano et al., 1997) (see also Fig. 5). The specificity of the existing Ras-effector loop mutations has been thoroughly tested in *Drosophila* imaginal discs (Prober and Edgar, 2002). Prober and Edgar show that RasV12[S35], able to interact with Raf in cell culture, can activate ERK/Rolled (via Raf) and induce Ras/Raf/ERK-specific transcriptional responses in wing and eye imaginal discs (figures 3,7 in Prober and Edgar) (Prober and Edgar, 2002). In contrast, RasV12[G37], unable to bind Raf in cell culture, cannot activate these responses, but is still capable of activating PI3K-specific read-outs (figures 3,7 in Prober and Edgar) (Prober and Edgar, 2002). Therefore we tested the Ras-effector loop mutations in constitutively activated RasV12 for their effects on ommatidial rotation.

Expression of RasV12 in developing photoreceptor precursor cells using common eye-specific drivers causes induction of many extra photoreceptors (Fortini et al., 1992), and thus does not allow the analysis of rotation (the orientation of individual ommatidial clusters cannot be determined unambiguously in the presence of extra photoreceptor cells). To circumvent this problem RasV12 and its effector loop isoforms were expressed in a limited photoreceptor subset after these have been determined as photoreceptors using the *m80.5-Gal4* driver (see Materials and methods for detail). RasV12 effects on photoreceptor number and fate were thus strongly reduced. To determine the full effect of activated Ras expression under the control *m80.5-Gal4*, we expressed constitutively active RasV12 that activates all known Ras-effectors. This gave rise to eyes with some gain and loss of photoreceptors and severe misrotations (Fig. 5A). The equivalent expression of RasV12[S35], thought to activate mainly Raf, also resulted in rotation abnormalities and occasional gain or loss of photoreceptors, again supporting a requirement of the Raf/MAPK cascade (Fig. 5B).

Strikingly, expression of RasV12[G37] and RasV12[C40] also caused misrotations, suggesting an involvement of additional Ras-effectors in this process. In particular, RasV12[G37], in which Raf activation is abolished (or at least strongly reduced), resulted in severe rotation defects (Fig. 5C,E), suggesting that PI3K, Rgl/Ral or Canoe might play a role in this cell motility process. Similarly, RasV12[C40], eliminating Raf activation, but maintaining weaker activation

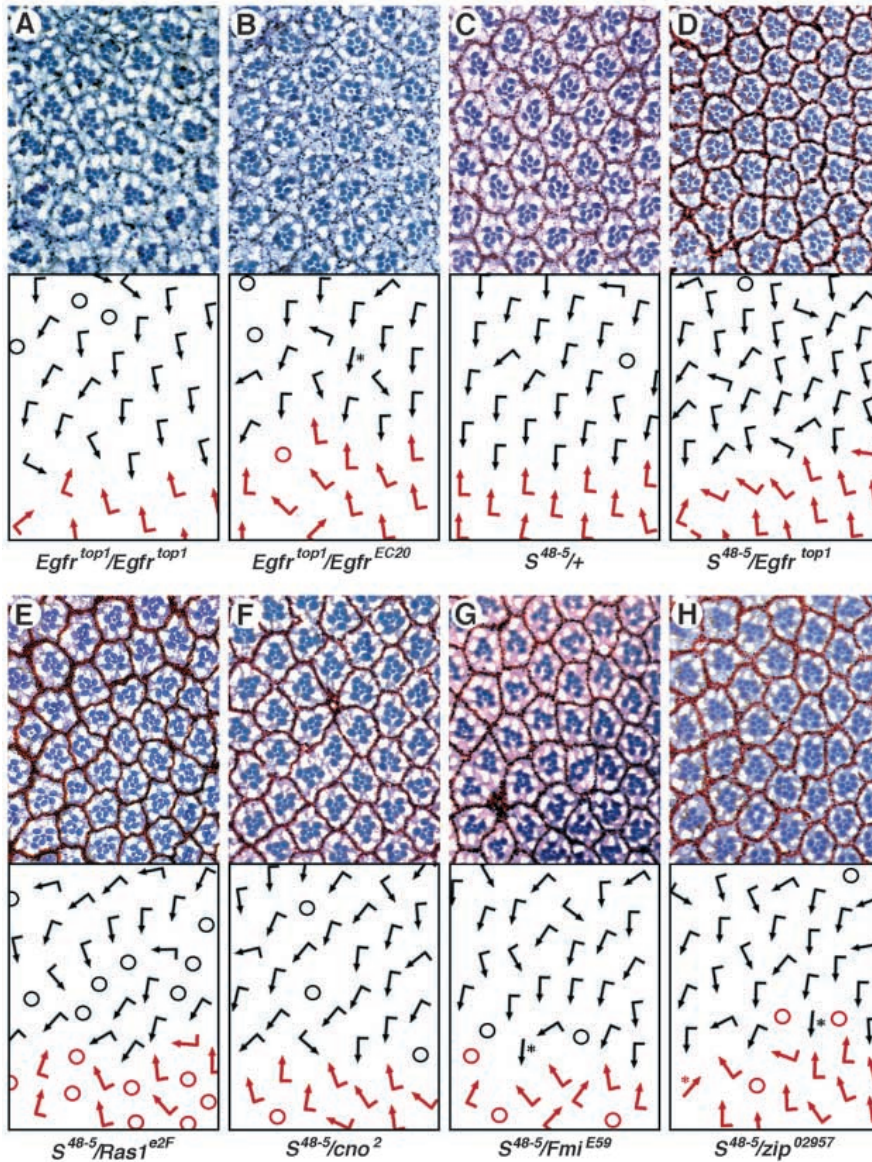


Fig. 3. Reduction in Egfr signaling affects ommatidial rotation. Panels show tangential adult eye sections around the equator and corresponding schemes reflecting ommatidial rotation. Orientation and arrows are as in Fig. 1, 'circles' represent ommatidia with loss of photoreceptors that cannot be unambiguously scored ('asterisk' indicates the rare event of a symmetrical ommatidium). (A) *Egfr^{top1}/Egfr^{top1}*, (B) *Egfr^{top1}/Egfr^{EC20}*. The hypomorphic *Egfr^{top1}* allele and *Egfr^{top1}/Egfr^{EC20}* allelic combination show typical rotation defects. Reduction of Egfr activity in *Star*^{-/+} causes rotation defects and is dominantly enhanced by components of Egfr signaling. *Star* is haplo-insufficient for eye development (note rare photoreceptor loss and rotation defects). The rotation defects are dominantly enhanced by removal of components of the Egfr/Ras pathway (compare C with D,E). Note that *canoe*, *flamingo* and *zipper* heterozygosity also enhances the *S*^{-/+} phenotype (see Table 1 for other genotypes and quantification). (C) *S⁴⁸⁻⁵/+*, (D) *S⁴⁸⁻⁵/+, Egfr^{top1}/+*, (E) *S⁴⁸⁻⁵/+; ras1^{e2F}/+*, (F) *S⁴⁸⁻⁵/+; cno2/+*, (G) *S⁴⁸⁻⁵/+; fmi^{E59}/+*, (H) *S⁴⁸⁻⁵/+; zipper⁰²⁹⁵⁷/+*.

of other effectors, also showed rotation abnormalities, albeit weaker than RasV12[G37] (Fig. 5D). Moreover, expression of RasV12[C40] under the control of the *sevenless* (*sev*) promoter (in R3/R4, R1/R6 and R7) resulted in strong rotation defects that were comparable to those seen with RasV12[G37] under *sev* control (Fig. 5E,F). Taken together, these data suggested an involvement of PI3K, Rgl/Ral or Canoe in ommatidial rotation.

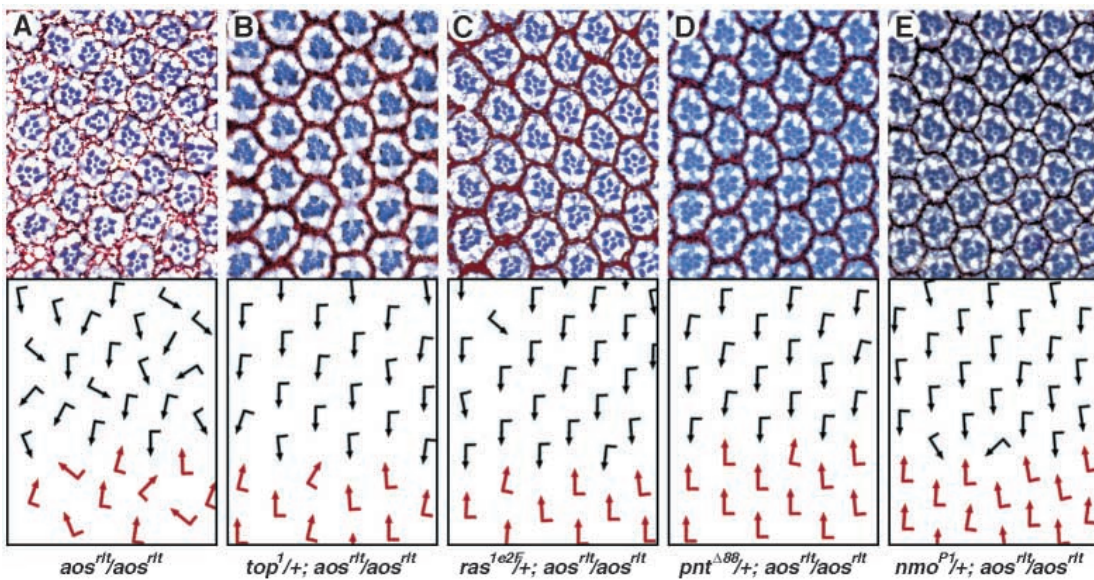


Fig. 4. *aosr^{lt}* is dominantly suppressed by components of the Egfr pathway and *nemo*. Panels show tangential adult eye sections through equatorial regions and corresponding schematic drawings indicating ommatidial rotation. Orientation and arrows are as in Fig. 1, 'stars' indicate rare event of symmetrical ommatidia. (A) *aosr^{lt}/aosr^{lt}*, (B) *Egfr^{top1}/+; aosr^{lt}/aosr^{lt}*, (C) *ras1^{e2F}/+; aosr^{lt}/aosr^{lt}*, (D) *pnt^{Δ88}/+; aosr^{lt}/aosr^{lt}*, (E) *nmo^{P1}/+; aosr^{lt}/aosr^{lt}*.

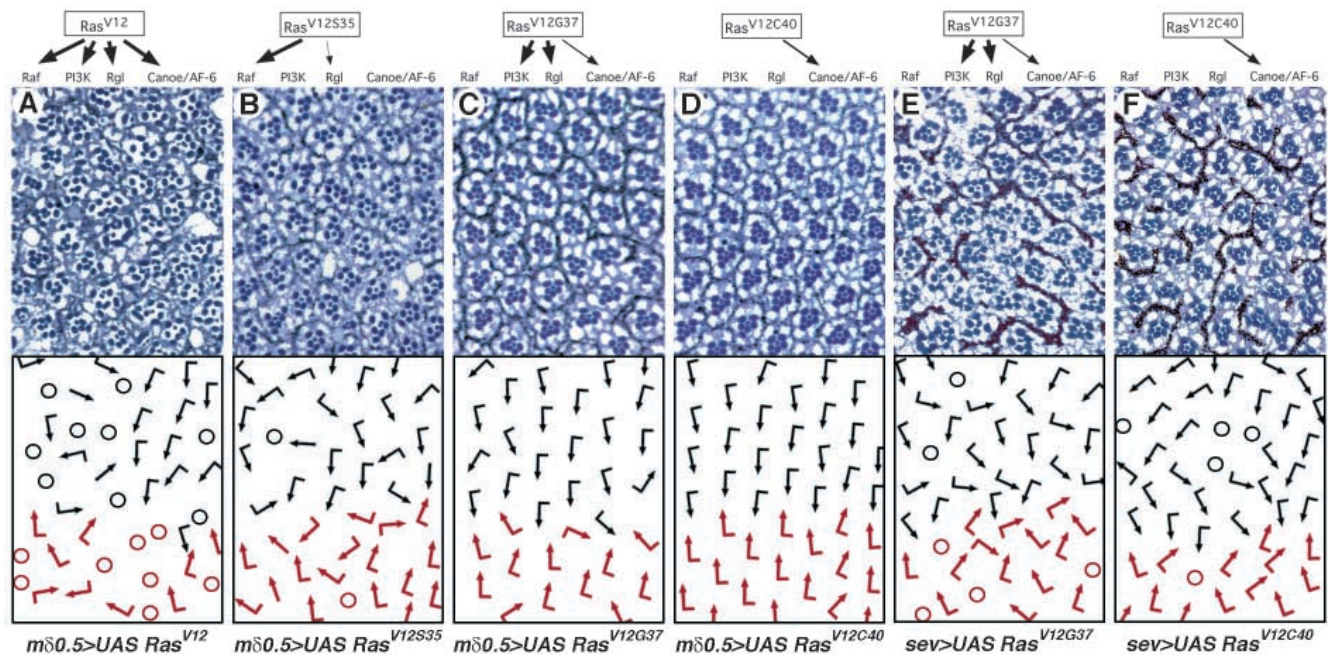


Fig. 5. Expression of Ras-effector loop mutations affects ommatidial rotation. (A-F) Tangential sections of adult eyes through equatorial regions and corresponding schematic drawings indicating ommatidial rotation. Orientation, arrows and circles are as in Figs 1, 3. (A) *mδ0.5Gal4>UAS-RasV12* (A similar, but weaker effect was generated by activated *Egfr* {λ-top} under *mδ0.5Gal4* control; not shown), (B) *mδ0.5Gal4>UAS-RasV12[S35]*, (C) *mδ0.5Gal4>UAS-RasV12[G37]*, (D) *mδ0.5Gal4>UAS-RasV12[C40]*, (E) *sevGal4>UAS-RasV12[G37]*, (F) *sevGal4>UAS-RasV12[C40]*.

Canoe is involved in ommatidial rotation

To confirm the RasV12[G37] effect and determine which of the three known effectors activated by RasV12[G37] is required in ommatidial rotation, we analyzed PI3K, Ral and Canoe directly. *UAS-PI3K* expressed under *mδ0.5-Gal4* had no effect on rotation (not shown), suggesting that PI3K is not required in this process. This is further supported by the lack of rotation defects in *dPI3K* mutant clones (Leevers et al., 1996). In contrast, expression of activated Ral (Sawamoto et al., 1999) as well as *mδ0.5-Gal4>UAS-Cno* exhibits rotation defects (Fig. 6A and not shown).

We next tested for a direct *canoe* (*cno*) requirement in ommatidial rotation using LOF alleles. First, we asked whether *cno* heterozygosity could interact with the *S^{48-5/+}* rotation phenotype. Strikingly, similar to the enhancement observed with *Egfr* or *Ras*, the *cno^{2/+}* and *cno^{3/+}* genotypes enhanced the *S^{48-5/+}* rotation phenotype (Fig. 3F, Table 1 and not shown).

cno is required for cone cell and photoreceptor differentiation and thus clones of null and strong alleles cause a general disorganization of the eye (Miyamoto et al., 1995) and are difficult to analyze for rotation defects. However, the hypomorphic *cno^{mis1}* allele is subviable in trans to the strong alleles *cno²* and *cno³* with mildly rough eyes, allowing an analysis of ommatidial rotation. Eye sections of such transheterozygous *cno* flies (e.g. *cno^{mis1/cno²}*) reveal severe rotation defects (Fig. 6B). To test whether such defects are already observed at the time when rotation takes place, we analyzed *cno* mutant third instar eye discs. Strikingly, rotation defects, comparable in strength to the stronger *aos* alleles, are apparent in *cno* eye imaginal discs (Fig. 2F). The discs were counterstained with anti-Elav (not shown) to ensure that the

photoreceptor complement is normal in such *cno* mutant discs and the observed rotation abnormalities are primary defects, which was indeed confirmed. A similar analysis of Ral/Rgl is precluded by the lack of suitable alleles. In summary, these data indicate that *cno* plays a critical role in ommatidial rotation and acts as an effector of Egfr/Ras signaling in this context (see also Discussion).

Effects of Egfr signaling on cell adhesion components during ommatidial rotation

As ommatidial rotation is a cell biological event, it is probable that among the main read-outs affected are cell-adhesion properties of the precluster cells and effects on cytoskeletal elements. This is further supported by our observations that (1) Raf/MAPK-independent and thus transcription-independent Egfr/Ras signaling pathways are important, and (2) that *canoe* is required in this context. To address this further, we performed two sets of experiments. First, we tested for genetic interactions between the dosage-sensitive *Star^{-/+}* rotation phenotype and selected factors required in cell adhesion and cytoskeletal regulation; and second, we directly analyzed whether cell-adhesion components such as cadherins and integrins are normally localized in *aos^{rlt}* and *cno^{Mis1}* mutant backgrounds.

To specifically test the involvement of cytoskeletal elements and adhesion as well as junctional components, we tested candidate genes for dominant interaction of the mild *Star* rotation phenotype (Table 1). This genetic data argue for an involvement of DE-Cadherin/*shotgun*, the atypical cadherin Flamingo (Fmi) (Usui et al., 1999), the adherens junction protein *canoe*, non-muscle myosin II (*zipper*), the septin

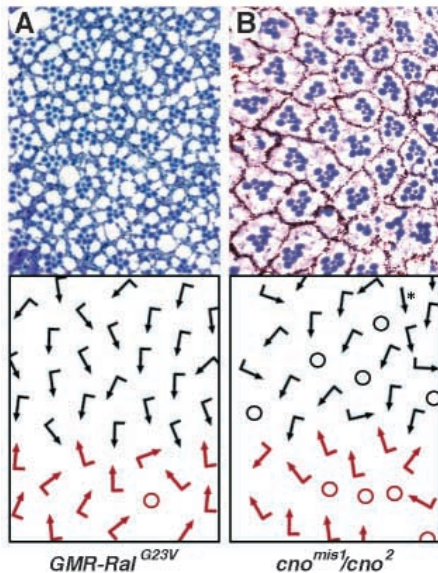


Fig. 6. Effects of Ral and Canoe on ommatidial rotation. (A,B) Tangential sections of adult eyes. An equatorial region is shown with the corresponding schematic presentation. Arrows and circles are drawn as in Figs 1, 3. (A) *GMR-Ral^{G23V}* (an activated isoform of Ral); (B) *cno^{mis1/cno²}*, a subviable hypomorphic allelic combination. Note the severe rotation defects in the *cno* mutant background.

peanut, and *capulet*, a protein with actin and adenylate cyclase-binding ability.

Next we examined the expression of Fmi and DE-cadherin in ommatidial preclusters during rotation. Strong LOF alleles of *Egfr* and its signaling components also affect cell proliferation, fate specification and survival, making the analysis of cell adhesion and junctional components in the context of rotation rather difficult. Thus we analyzed localization of the cadherins and Arm/ β -catenin in imaginal discs of the rotation-specific *aos^{rlt}* allele.

Although the overall expression and localization of DE-Cadherin and Arm/ β -catenin are largely unaffected (Fig. 7A,A'',B,B'',C,C'',D,D'' and not shown), the localization of Fmi is changed in *aos^{rlt}* discs (Fig. 7A,A',B,B',G,H). In WT, Fmi is initially present apically in all cells of the morphogenetic furrow and subsequently becomes asymmetrically enriched in the R3/R4 precursor pair (column 4-5, Fig. 7A,C,G) (Das et al., 2002). In and posterior to column 6 Fmi is expressed at the membrane of R4, and largely depleted from R3 membranes that do not touch R4, forming a horseshoe-like R4-specific pattern (Fig. 7A,A',G). In contrast, in *aos^{rlt}* discs Fmi restriction to the R4 precursor is generally delayed, and often not established even in columns 8-12, where high levels of Fmi are still seen around the apical membrane cortex of R3 and R4 (Fig. 7B,H). As Fmi is thought to act as a homophilic cell-adhesion molecule (Usui et al., 1999), its increased presence on R3 membranes should have a direct effect on Fmi localization in neighboring cells and thus possibly the adhesive properties of the precluster. It is worth noting that although Fmi is required during PCP establishment and R3/R4 cell-fate specification, the delay in Fmi restriction to R4 has no significant effect on the R3/R4 cell-fate decision.

Although Fmi interacts with Fz and Notch in this context, the R4-specific *m δ -lacZ* marker does not differ significantly from WT (Fig. 7E,F) and adult *aos^{rlt}* eyes also display no defects in R3/R4 specification. Thus, it appears that the delay in Fmi localization specifically affects ommatidial rotation, probably through adhesion, and possibly explains the broad range of rotation angles in *aos^{rlt}* and other *Egfr* pathway mutants.

Taken together with the genetic interactions, these data suggest that *Egfr* signaling impinges on the cell adhesive properties of ommatidial clusters during the rotation process (see below).

Discussion

Our data demonstrate that *Egfr* signaling regulates ommatidial rotation. Loss and gain-of-function data indicate that at least two Ras-effector pathways, the Ras/Raf/MAPK cascade and Ras/Cno signaling, are required downstream of *Egfr* in this context of cellular motility (Fig. 8). We provide further evidence for an involvement of cadherins and non-muscle myosin II as downstream components controlling rotation. In particular, the involvement of the cadherin Fmi, a PCP gene, downstream of *Egfr* signaling, is of interest as it provides the first link between PCP establishment and *Egfr* signaling.

The role of *Egfr* in rotation

What is the role of *Egfr* signaling in the regulation of ommatidial rotation? It is of interest to note that both loss and gain-of-function phenotypes of *Egfr* and its signaling pathway components result in very similar phenotypes, with ommatidia being both under- and overrotated. This is in contrast to *nmo* and *sca* mutants, in which ommatidia generally underrotate (*nmo*) or overrotate (*sca*). Although *rlt* has been implicated to stop rotation at 90° (Choi and Benzer, 1994), our data suggest a general role for *Egfr* in this process affecting more than just its termination. The observed underrotation in *Egfr* LOF genotypes, such as *Egfr^{top1}* alleles or *Star* heterozygosity, is consistent with a positive requirement of *Egfr* in the process. The presence of overrotated ommatidia in these genotypes questions this simple interpretation. However, the overrotations could be explained by the feedback loops that regulate the levels of *Egfr* signaling. An initial reduction of *Egfr* signaling, such as in the *Egfr^{top1}* or *Star* backgrounds, also results in a reduced induction of the negative feedback loop (mediated through *aos*), subsequently leading to a mild increase in *Egfr* signaling. Thus, as null mutations of *Egfr* cannot be easily analyzed in this context, it is possible that in all genotypes analyzed the absolute levels of *Egfr* signaling fluctuate, causing the observed phenotypic variations. In this context, it is noteworthy that the gain-of-function *Egfr* mutation *Elp^{B1}* exhibits only very minor rotation abnormalities, but significant photoreceptor loss because of overproduction of Aos (Lesokhin et al., 1999).

An alternative interpretation of the observed *Egfr* rotation defects could be a general regulatory input from *Egfr* signaling into the process of rotation. *Egfr*/Ras signaling could serve as a 'gas' or 'brake' pedal throughout the process, thus regulating the strength of another signaling input. This is supported by the observations that ommatidia can be significantly overrotated before the end of the process, suggesting a continuous *Egfr* input.

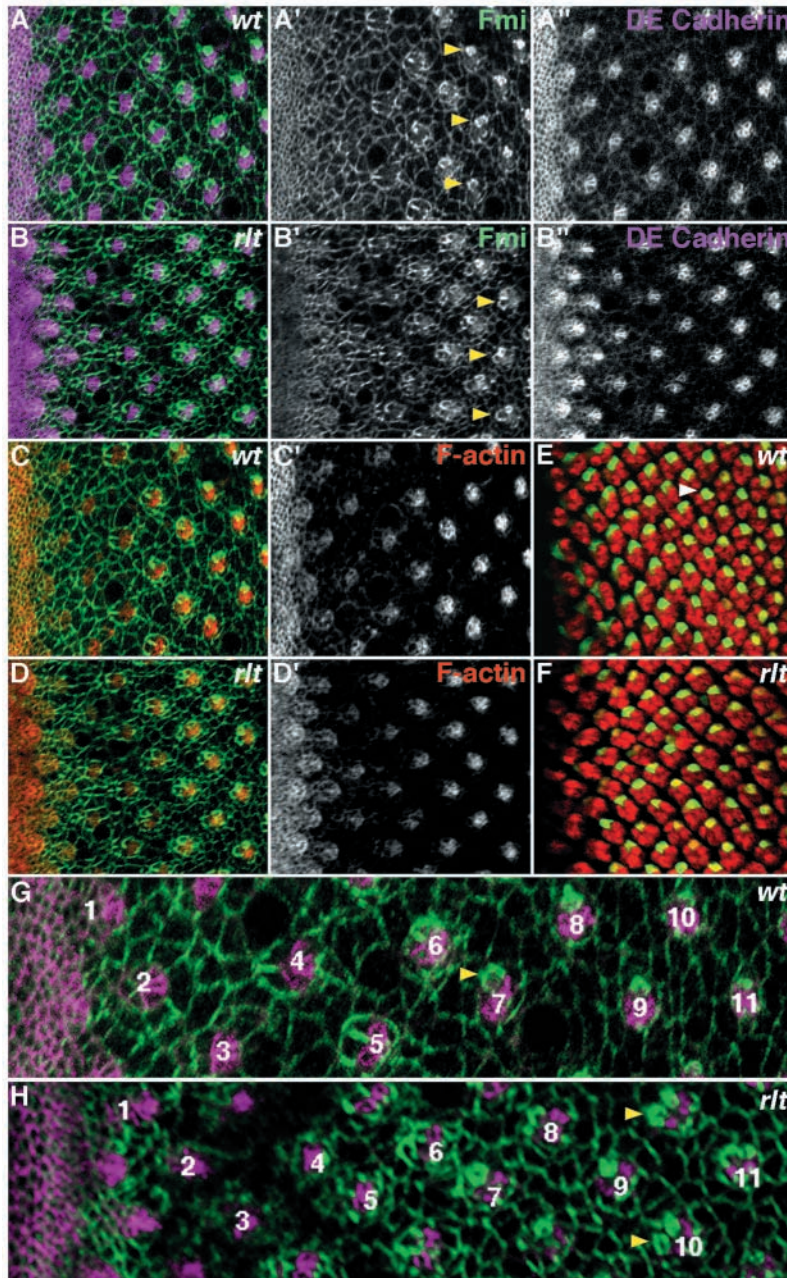


Fig. 7. Expression of the atypical cadherin Flamingo is altered within the R3/R4 precursor pair in *argos*^{rlt} discs. Confocal images of third instar eye discs are shown, anterior is left. The morphogenetic furrow is at the left edge of each panel. (A,C,G) wild-type (WT) and (B,D,H) *aos*^{rlt} discs stained with anti-Fmi (green in A-D, black and white in A' and B'), anti-DE-cadherin (magenta in A,B and b/w in A'' and B'') and F-actin (red, labeled by Phalloidin, in C,D and b/w in C',D'). A higher magnification of the Fmi and DE-cadherin stainings in similar discs is shown in G and H (ommatidial rows 1-11). Note that Fmi, in WT downregulated in the R3 precursor by row 7 and subsequently detected only around R4, is still present in R3 cells in *aos*^{rlt} until rows 11-12 and beyond. Typical WT and mutant examples are highlighted with yellow arrowheads in A',B',G,H. Abnormal localization of Fmi does not correlate with R3/R4 cell-fate defects, as the R4-specific marker *mδ-lacZ* is restricted to R4 in WT (E) and *aos*^{rlt} mutant (F) preclusters.

Nevertheless, the phenotypes observed in third instar eye imaginal discs suggest that the first 45° rotation is much less affected than the second step. This indicates that Egfr signaling plays an important role in the second 45° turn, and serves a lesser role in the first rotation step (see also Brown and Freeman, 2003). In summary, we suggest that if Egfr signaling is deregulated, either through a reduction in *aos* or *Egfr* function, rotation is disturbed during the second 45° step. We propose that the restriction of Egfr signaling, required for correct rotation, is mediated by an autoregulatory feedback loop via *aos*. In parallel, the mechanistic cell motility aspects of rotation are probably regulated at least in part via the Ras effector Cno.

Ommatidial rotation and cell adhesion factors

The enhancement of the rotation defects in *S* heterozygous eyes through several cell adhesion molecules and cytoskeletal regulators implicates Egfr signaling in controlling the mechanistic 'cell-motility process'. Our results, showing that expression of the cadherin Fmi is altered in *aos*^{rlt} eye discs, suggest that Egfr signaling impinges on cell-adhesion properties of the ommatidial precluster. It

is interesting to note that in WT the expression of DE-cadherin and Fmi are almost exclusive within the early precluster: membranes between the R8, R2 and R5 cells show high levels of DE-cadherin, whereas the membranes of the R3/R4 precursors show high Fmi levels (Fig. 7A,A'',B,B'',G,H). Thus

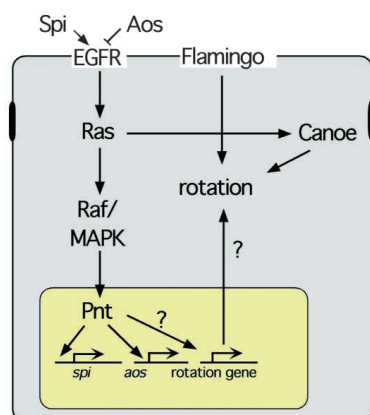


Fig. 8. Model for the role of Egfr/Ras signaling in the context of ommatidial rotation. Raf/MAPK nuclear signaling to Pnt upregulates the transcription of activating (*spi*) and inhibiting (*aos*) Egfr ligands and might regulate transcription of additional genes important for ommatidial rotation. Egfr/Ras signal via Canoe to affect cytoskeletal and/or cell junctional elements (junctions are schematized as black oval regions on membrane). In addition, Egfr signaling affects Flamingo localization (the effector regulating this aspect is not known).

the regulation of such differences among distinct cadherin proteins might be important for normal ommatidial rotation to occur. The input of Egfr signaling on cell adhesion is further supported by the genetic interactions of *Star* with *fmi* and DE-cadherin/*shotgun*.

These analyses of the regulation of cell adhesion molecules in the context of ommatidial rotation will help to start unravel the mechanistic aspects of rotation.

The role of Canoe in ommatidial rotation

The Egfr/Ras/Cno link is intriguing for several reasons. The *cno* gene was originally identified as a mutation affecting the dorsal closure process during embryogenesis (Jürgens, 1984; Takahashi et al., 1998). Cno shows a genetic and molecular link to Ras: it contains two Ras-interacting domains and binds both WT Ras and activated Ras-V12 (Kuriyama et al., 1996; Matsuo et al., 1997). In addition, Cno has been postulated to link cytoskeletal elements to cellular junctions via its ability to bind actin (Mandai et al., 1997), its interaction with ZO-1/Pyd and its homology with kinesin and myosin-like domains (Takahashi et al., 1998). Thus Cno could directly mediate an Egfr/Ras signal to cytoskeletal and cell architecture elements through its association with adherens junctions and its kinesin and myosin-like domains (Miyamoto et al., 1995). Interestingly, Zipper does not only show a similar interaction with *Star*, like Cno (Table 1), but it is also required during embryonic dorsal closure (Young et al., 1993), and thus a more general Cno-Zipper link might exist in cell motility contexts.

A second interesting feature of *cno* is that it has been genetically linked to *sca* and *Notch* signaling (Miyamoto et al., 1995). First, the phenotype of the *sca*¹ allele is strongly enhanced by *cno*^{-/+}. Second, *cno* alleles also display *Notch*-like phenotypes in the wing and a GOF *Notch* allele, *Notch*^{Abruptex}, is suppressed by *cno* (Miyamoto et al., 1995). Although the biochemical role of Sca remains obscure, it has been linked to Notch, possibly as a Notch ligand, in several contexts (Powell et al., 2001). Thus as *sca* has recently been implicated in ommatidial rotation (Chou and Chien, 2002), the link between Cno and Sca/Notch is intriguing. Taken together, Cno could serve as a factor integrating signaling input from different pathways, e.g. Egfr and Notch in this process, and relaying this to cytoskeletal elements. The Canoe link is also interesting from a disease point of view as its human homologue AF6 is the critical partner of ALL1 in a chimeric protein associated with myeloid leukemia (Prasad et al., 1993). Thus taken together, Cno could serve as a factor integrating signaling input from different pathways, e.g. Egfr and Notch in ommatidial rotation, and relaying this to the regulation of cell adhesion and cytoskeletal elements in the context of a developmental patterning process or disease.

Concluding remarks

We demonstrate that Egfr/Ras signaling plays a general role in the regulation of ommatidial rotation. We further identify Canoe as an effector of Ras in this context. Although much was known about how ommatidial chirality and the associated R3/R4 cell-fate decision are regulated (Fz/PCP-Notch signaling), no clear link between the mechanistic aspects of ommatidial rotation and Fz/PCP signaling previously existed. We show a first link between Egfr signaling and PCP genes, namely *Fmi*. A further connection between Egfr signaling and PCP establishment is provided by Zipper, which acts

downstream of Fz/Dsh and Rok in wing PCP (Winter et al., 2001) and modifies the *Star* rotation phenotype. The identification of the Egfr pathway and its regulation of Fmi/cadherin-mediated cell adhesion will serve as an important entry point to further such studies.

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